

# Kenneth Ly

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## EDUCATION

**California State University, Fullerton (CSUF)**

**Fullerton, CA**

*B.S. Computer Science*

*Expected Graduation: May 2027*

- **Cumulative GPA:** 3.64
- **Activities & Societies:** Association of Computing Machinery
- **Coursework:** Data Structures, Object-Oriented Programming, Computer Assembly and Organization, Operating System Concepts, File Structure and Database, Software Engineering, Compilers and Languages

## PROJECTS

### Sonar Scanner

*C++, Arduino UNO*

- Engineered a radar-style object detection system by programming an Arduino UNO in C++ to control an ultrasonic sensor mounted on a servo motor sweeping 180°, visualizing detected distances on a live radar interface.
- Implemented real-time object mapping using ultrasonic sonar data processed via Arduino IDE and servo-driven angular sweeps, producing an on-screen radar visualization of surrounding obstacles.
- Leveraged AI-assisted development to optimize sensor sweep logic and data handling algorithms, reducing measurement latency by 25% and improving distance reading accuracy to 3cm precision for enhanced radar display clarity.

### Clean Water Station Locator (Hackathon Project)

*HTML, CSS, JavaScript, Leaflet.js*

- Collaborated in a 3-person team to design and build a web application within a 24-hour hackathon timeframe.
- Developed an interactive map-based interface to guide CSUF students to nearby clean water stations using geolocation data.
- Integrated the leaflet.js mapping library to display location markers and dynamically update results based on user position.

### CodePath Tech Exchange Full-Stack Application (CTEx)

*BigQuery, Google Cloud, Python, SQL, Docker*

- Collaborated in a 4-person team to design, build, and deploy a full-stack web application using Docker, Google Cloud Platform, BigQuery, and Vertex AI, integrating third-party APIs for core functionality
- Ran weekly agile sprints, using GitHub to track progress and assign tasks via a point-based system weighted by task difficulty
- Led weekly stand-up meetings to review sprint progress, problems to address, and plan the next steps, strengthening team communication and my project management skills

## TECHNICAL SKILLS

- **Languages:** C++, Python, JavaScript, HTML, CSS, x86 Assembly
- **Tools & Methodologies:** Git, React, Linux, Arduino, Agile, Scrum, Sprints

## ADDITIONAL INFORMATION

- Eligible and willing to undergo background investigation and security clearance process
- U.S. work authorized